Physics - Graduate Level Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. 10 calories of heat are supplied to 1 g of water. How much does the mass of the water increase?

   (a) $5.5 \times 10^{-}13g$

   (a) 2.5 × 10-13 g
   (b) $2.0 \times 10^{-}12g$
   (b) 4.7 × 10-13 g
   (c) $8.6 \times 10^{-}13g$

2. Compute the mean molecular speed v in the light gas hydrogen (H 2) in m/s

   (a) 1750.0
   (b) 1500.0
   (c) 2200.0
   (d) 2100.0
   (e) 1400.0

3. Consider two mercury barometers, one having a cross-sectional area of 1 cm 2 and the other 2 cm 2. The height of mercury in the narrower tube is

   (a) two-thirds
   (b) twice
   (c) the same
   (d) half
   (e) four times higher

4. Which one of these constellations is not located along the Milky Way in the sky?

   (a) Virgo
   (b) Perseus
   (c) Ursa Major
   (d) Sagittarius
   (e) Aquila

5. An object, starting from rest, is given an acceleration of 16 feet per $second^2$ for 3 seconds. What is its speed at the end of 3 seconds?

(a) 48 ft per sec
(b) 64 ft per sec
(c) 30 ft per sec
(d) 72 ft per sec
(e) 16 ft per sec

True / False (10 marks)

Answer True or False

6. True or False: The angular magnification of the refracting telescope is 10, given the focal length of the eye-piece lens and the separation distance.
7. True or False: The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/c is approximately 1.0 GeV/c^2.
8. True or False: The efficiency of a device can exceed 100% if it utilizes energy from multiple sources.
9. True or False: A 2 mm thick neutral density filter that originally transmits 70% light at 0.1 mm will transmit approximately 0.08% light.
10. True or False: Doubling the volume of an object with constant mass will cause its density to triple.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the shape of the illuminated area on a screen at distances of 1 cm and 2 m from a 2 mm $\times$ 2 mm plane mirror reflecting sunlight, providing a detailed analysis of the underlying principles.
12. A converging lens with a focal length of 5.0 cm is used as a simple magnifier to produce a virtual image located 25 cm from the eye. Determine the object distance from the lens and calculate the magnification. Discuss the underlying principles and formulas used in your analysis.

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